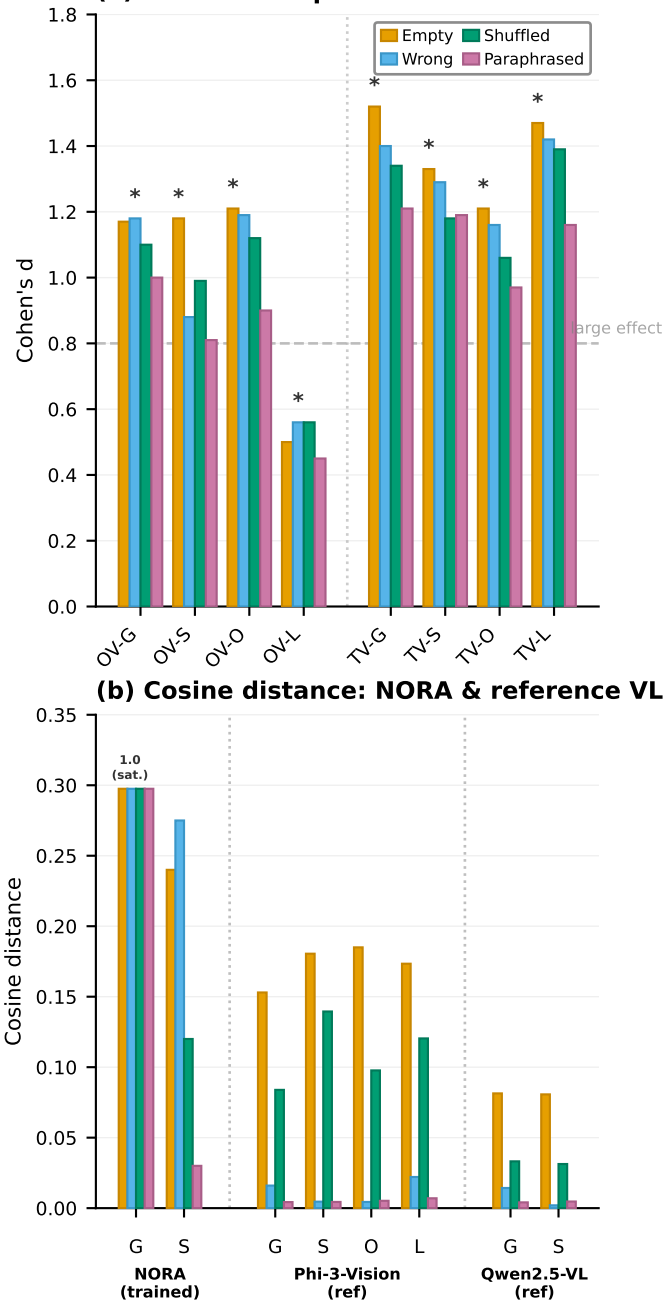
(a) Cohen's d: OpenVLA & TracVLA
Empty
Shuffled
Wrong
Paraphrased
Cohen's d
1.8
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
large effect
OV-G
OV-S
OV-O
OV-L
TV-G
TV-S
TV-O
TV-L
(b) Cosine distance: NORA & reference VLMs
Cosine distance
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
1.0
(sat.)
G
S
NORA
(trained)
Phi-3-Vision
(ref)
Qwen2.5-VL
(ref)